# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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## CLASS 10 — SCIENCE · HARD LEVEL UNIT TEST SET 2

Chapters 1–4: **Chemical Reactions and Equations, Acids, Bases and Salts, Metals and Non-metals, Carbon and its Compounds** | RBSE Pattern | Session 2026–27

**Time: 2 Hours** Name: _____ Roll No.: _____ **Maximum Marks: 50**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

## SECTION — A (Objective Type Questions) 15 × 1 = 15

*Choose the correct option and write it in your answer book.*

**1.** Assertion (A): Unsaturated hydrocarbons burn with a sooty flame. Reason (R): They have a higher percentage of carbon so burn incompletely. 1

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**2.** Mg burns in oxygen: 24 g Mg gives 40 g MgO. Mass of oxygen used is: 1

**(a)** 16 g **(b)** 40 g **(c)** 24 g **(d)** 64 g

**3.** 20 mL of NaOH is neutralised by 10 mL of HCl. To neutralise 50 mL of the same NaOH, the volume of HCl needed is: 1

**(a)** 10 mL **(b)** 25 mL **(c)** 100 mL **(d)** 5 mL

**4.** Four metals P, Q, R, S: P displaces Q from its salt; Q displaces R; R displaces S. The least reactive is: 1

**(a)** Q **(b)** S **(c)** R **(d)** P

**5.** In the balanced equation $2\,C_2H_6 + 7\,O_2 \longrightarrow 4\,CO_2 + 6\,H_2O$, the coefficient of water is: 1

**(a)** 3 **(b)** 6 **(c)** 7 **(d)** 4

**6.** A student places metal M in blue copper sulphate solution. The blue colour fades to colourless and a reddish-brown solid forms. M could be: 1

**(a)** zinc **(b)** mercury **(c)** gold **(d)** silver

**7.** A salt X, when heated, loses water and turns from blue to white. X is: 1

**(a)** $CuSO_4 \cdot 5\,H_2O$ **(b)** $FeSO_4 \cdot 7\,H_2O$ **(c)** $CaSO_4 \cdot 2\,H_2O$ **(d)** $Na_2CO_3 \cdot 10\,H_2O$

**8.** Green copper carbonate on heating gives black solid X. X, when heated in hydrogen, gives a brown metal. The metal is: 1

**(a)** zinc **(b)** copper **(c)** iron **(d)** lead

**9.** An organic compound Y gives effervescence with $NaHCO_3$ and is used as a preservative. Y is: 1

**(a)** ethanoic acid **(b)** methanal **(c)** ethane **(d)** ethanol

**10.** Assertion (A): Chips packets are flushed with nitrogen. Reason (R): Nitrogen is a reactive gas that speeds up oxidation of oil. 1

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**11.** Assertion (A): Aqueous solutions of glucose and alcohol do not conduct electricity. Reason (R): They do not ionise in water to give $H^+$ ions. 1

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**12.** Assertion (A): Ionic compounds are brittle. Reason (R): In ionic crystals, like ions can come near each other on a slight shift, causing repulsion. 1

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**13.** **Assertion (A):** Carbon forms a very large number of compounds. 1
**Reason (R):** Carbon is tetravalent and shows the property of catenation.
Choose the correct option:

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**14.** **Assertion (A):** A white precipitate is formed when barium chloride solution is mixed with sodium sulphate solution. 1
**Reason (R):** The reaction between barium chloride and sodium sulphate is a double displacement reaction.
Choose the correct option:

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**15.** **Assertion (A):** Bleaching powder is used for disinfecting drinking water. 1
**Reason (R):** Bleaching powder is produced by the action of chlorine on dry slaked lime.
Choose the correct option:

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

## SECTION — B (Very Short Answer Type) 5 × 2 = 10

*Answer in about 20–30 words each.*

**16.** How much gypsum (172 g) heated gives how much plaster of Paris? Give the mass of water lost. 2

**17.** Why are metals in the middle of the reactivity series usually obtained from their oxides rather than sulphides or carbonates? 2

**18.** An alkane has 22 covalent bonds. Find its molecular formula. 2

**19.** Write the balanced equation for the thermal decomposition of zinc carbonate and name the gas evolved. 2

**20.** Why is $H_2SO_4$ never diluted in a glass bottle by pouring water on it? 2

## SECTION — C (Short Answer Type) — 5 × 3 = 15

*Answer in about 40–60 words each.*

**21.** Determine the number of C-C and C-H bonds in butane and in isobutane and comment. 3

**22.** Dilute hydrochloric acid is added to zinc granules. Name the gas, state the test for it and write the equation. 3

**23.** Explain why curd should not be kept in copper or brass vessels but tomato ketchup can be stored in glass. 3

**24.** Explain why ionic compounds like NaCl conduct electricity in aqueous solution while sugar solution does not. 3

OR

State the effect of chain length on physical state in the alkane series with examples. 3

**25.** Balance $Fe_3O_4 + H_2 \longrightarrow Fe + H_2O$ and name the substance oxidised. 3

OR

A farmer's soil has pH 5. Name one chemical he can add, why, and the effect on crop. 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**26.** **Case study.** A gardener, Harpal, tested four soil-water samples with a pH paper. Sample P showed pH 5, sample Q pH 7, sample R pH 9 and sample S pH 13. 5

(i) Which sample is neutral?
(ii) Which sample is the strongest base?
(iii) Which sample(s) will turn red litmus blue?
(iv) Harpal wants to grow a crop in the acidic sample P. Name a substance he can add to the soil to bring the pH nearer to 7 and give the reason.

**27.** Explain: 5

(a) why sodium is kept in kerosene
(b) why aluminium foil is used in packaging
(c) why gold is alloyed
(d) why ionic compounds are brittle.

OR

Draw structures and name all isomers of $C_4H_{10}$, $C_4H_8$ positions (but-1-ene, but-2-ene) and explain the difference between structural isomers and homologues. 5

## END OF QUESTION PAPER — Total: 50 Marks